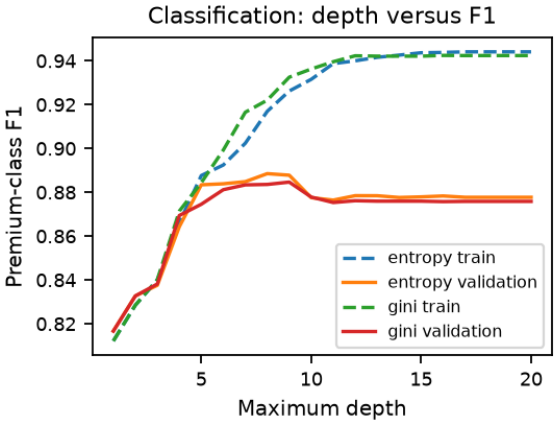
Decision Tree built from scratch

- NumPy tree for log-price regression and premium/standard classification.
- Sort each feature; test midpoints of adjacent distinct values.
- Maximize impurity decrease: MSE, Gini, or entropy.
- Four growth controls: depth, split size, leaf size, and minimum gain.
- Leaf output: mean target or class frequencies.



Validation selected entropy, depth 8, and at least 20 samples per leaf.

Tree results and failure modes

Held-out metric	Our tree	Matched sklearn tree
Price RMSE (AZN)	389,483	389,485
Price MAE (AZN)	103,067	103,057
Tier F1	0.8853	0.8853
Tier ROC-AUC	0.9513	0.9513

- Same 5,000 train rows; validation and test use full splits (9,406 each).
- Tier errors: 42.1% near the 212,000 AZN boundary, versus 11.3% overall.
- Price MAE: 566,294 AZN for luxury listings; 714,521 AZN for land plots.

This is a single held-out split. The team benchmark supplies cross-validation uncertainty.